Fowler (2024) on the Japanese commuting zones

August 26, 2026

What this deck reports

- Every table and figure of Fowler (2024) that the Japanese data supports, in the order that paper presents them.
- Delineations for 1980 to 2020; this deck shows the 2010 against 2020 comparison, the same decade spacing as the source paper.
- The wage-correlation block, that paper's Figure 2 and the Connection rows of its Table 2, waits on the Basic Survey on Wage Structure microdata.
- Every body slide carries a button to the appendix slide that shows all nine census years.

Source paper: `https://doi.org/10.1038/s41597-024-03829-5`

The delineation

For municipalities i and j with commuting flows f_{ij} and resident labour forces W_i, the proportional flow is

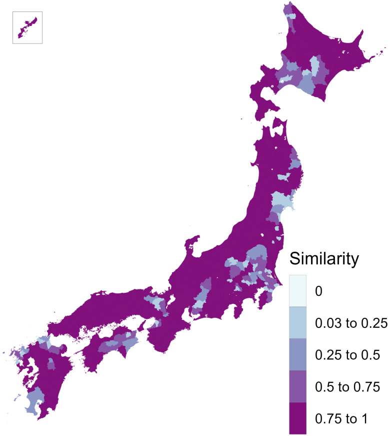
$$p_{ij} = \frac{f_{ij} + f_{ji}}{\min(W_i, W_j)}, \qquad p_{ij} \leq 0.999,$$

and the dissimilarity is $1 - p_{ij}$ with a zero diagonal.

- Average-linkage hierarchical clustering, cut at a fixed tree height of 0.977.
- Commuting matrix: residents mainly working, ordinary households, the only definition available for 2020.
- 1,656 municipalities: the four main islands and the Okinawa main island, plus every municipality joined to one by a permanent road link. The twenty-three special wards of Tokyo count as one, as the ordinance-designated cities do.
- Cores and urban areas for the fit statistics: the Urban Employment Area central cities and their areas.

Similarity between the 2010 and the 2020 delineations

Figure 1 of Fowler (2024)



Jaccard similarity: the overlap of a municipality's two commuting zones over their union. Mean 0.85.

▸ Every comparison pair

Diagnostic statistics

Table 1 of Fowler (2024)

	2010	2020
Municipalities in the delineation	1,656	1,655
Commuting zones	231	223
Single-municipality zones	9	10
Municipalities in the largest zone	27	29
Non-contiguous zones	0	1
Smallest zone, resident labour force	530	475
Average zone, resident labour force	202,694	207,532
Largest zone, resident labour force	4,929,463	5,624,897
Smallest zone area (sq. km)	184	100
Average zone area (sq. km)	1,574	1,630
Largest zone area (sq. km)	5,744	5,268
Compactness of zones	0.30	0.30

Compactness is the Polsby–Popper ratio, $4\pi A/P^2$ of the dissolved zone, averaged across zones.

▸ All nine census years

Fit statistics

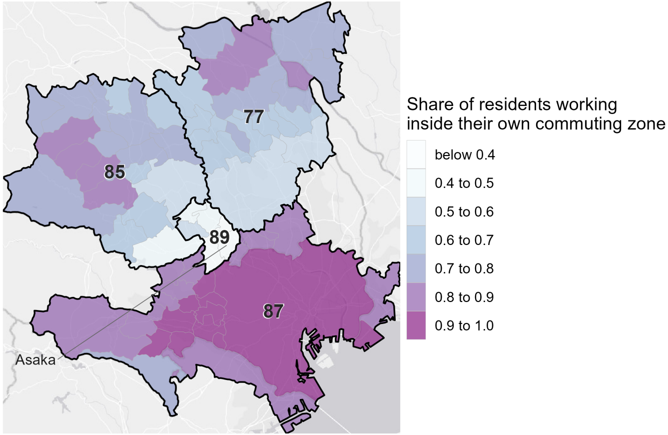
Table 2 of Fowler (2024)

	2010	2020
Core		
Urban areas split across zones	31	33
Zones containing a core	72.7%	67.3%
Residents working in a core of their zone	52.3%	51.9%
Workforce living in a core of their zone	47.9%	47.7%
Connection		
Minimum wage correlation	pending	
Mean wage correlation	pending	
Mean wage correlation, multi-municipality zones	pending	
Containment		
Minimum contained	44.5%	40.5%
Mean contained	90.0%	90.4%
Share of the labour force contained	85.8%	86.1%

All nine census years

The commuting zone with the lowest mean containment

Figure 3 of Fowler (2024)



Basemap: Esri, HERE, Garmin and OpenStreetMap contributors

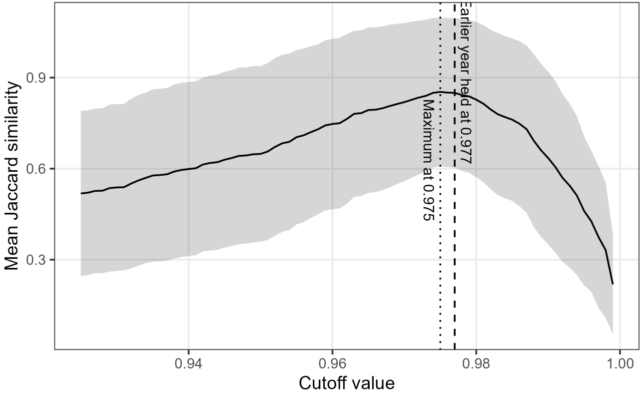
Six municipalities in southern Saitama at 0.41, between the Tokyo zone and the zones beyond.

Technical validation

- The cutoff. It is inherited rather than derived. Does the similarity between delineations pick out a better value?
- Non-contiguity. Zones that hold a municipality sharing its zone with none of its neighbours.
- Split urban areas. Externally defined urban areas cut across more than one commuting zone.

Similarity as a function of the cutoff

Figure 4 of Fowler (2024)

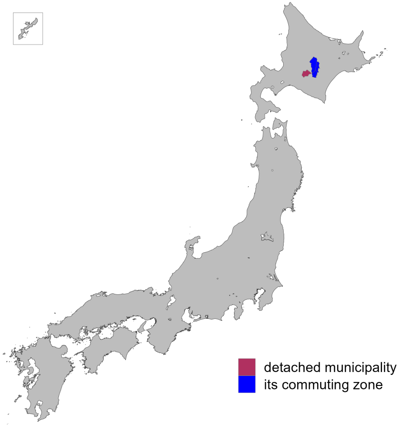


Holding 2010 at 0.977, the peak is 0.85 at 0.975, and the curve stays within 0.005 of it from 0.974 to 0.977, spanning 236 down to 223 zones.

▸ The two cutoffs compared

Non-contiguous municipalities

Figure 5 of Fowler (2024)



One municipality in 2020, Shimukappu in Hokkaido, 991 metres from the nearest part of its own zone.

Reassignments that would remove non-contiguity

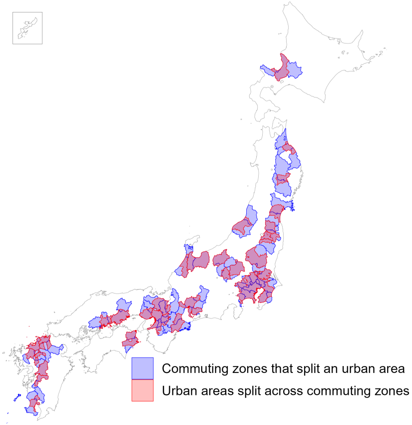
Table 3 of Fowler (2024)

Municipality	Zone	To own zone	Nearest zone	To that zone	Suggestion
Shimukappu	26	95.9%	18	2.9%	Retain

Its own zone takes 96 percent of its commuters, the adjacent zone 3 percent.

Urban areas split across commuting zones

Figure 6 of Fowler (2024)



33 of the 208 Urban Employment Areas in 2020, concentrated along the Pacific corridor.

All nine census years

Appendix

Diagnostic statistics, 1980 to 2020

Year	Municipalities	Commuting zones	Single-municipality	Largest zone	Non-contiguous
1980	1,656	384	71	31	2
1985	1,656	341	53	29	2
1990	1,656	307	34	28	1
1995	1,656	281	24	28	2
2000	1,656	261	15	30	1
2005	1,656	239	12	27	0
2010	1,656	231	9	27	0
2015	1,651	225	9	29	2
2020	1,655	223	10	29	1

Counts at the 0.977 cutoff. Non-contiguous zones are those holding a municipality with no neighbour in its own zone.

Zone size and shape, 1980 to 2020

Year	Resident labour force			Area (sq. km)			Compactness
	Smallest	Average	Largest	Smallest	Average	Largest	
1980	286	120,319	4,747,659	16	947	3,464	0.33
1985	650	140,701	5,265,234	24	1,066	3,839	0.32
1990	788	168,711	5,672,523	24	1,184	4,429	0.32
1995	723	189,749	5,312,186	24	1,294	4,429	0.32
2000	523	202,706	6,000,065	109	1,393	4,429	0.31
2005	295	211,525	5,355,045	109	1,521	4,733	0.31
2010	530	202,694	4,929,463	184	1,574	5,744	0.30
2015	479	205,844	4,819,056	100	1,613	4,909	0.30
2020	475	207,532	5,624,897	100	1,630	5,268	0.30

Population is the resident labour force carried in the commuting matrix. Compactness is the Polsby–Popper ratio.

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Containment, 1980 to 2020

Year	Minimum	Mean	Labour-force weighted mean	Share of the labour force	Mean work contained
1980	43.6%	91.1%	88.7%	88.7%	94.5%
1985	23.0%	90.4%	88.3%	88.1%	94.0%
1990	21.2%	89.9%	87.6%	87.1%	93.4%
1995	20.9%	89.2%	86.7%	86.3%	92.6%
2000	40.6%	89.9%	87.0%	86.6%	92.3%
2005	44.7%	90.1%	86.7%	86.1%	92.3%
2010	44.5%	90.0%	86.5%	85.8%	92.1%
2015	39.0%	90.1%	86.1%	85.3%	91.5%
2020	40.5%	90.4%	87.0%	86.1%	91.4%

The minimum and the mean are taken over commuting zones; the share of the labour force is a single national ratio.

▸ Core measures ◂ Back

Core measures, 1980 to 2020

Year	Urban Employment Area				District population 10,000 or more		
	With core	Work in	Live in	Split	With core	Work in	Live in
1980	55.7%	58.1%	53.5%	50	60.7%	74.0%	72.5%
1985	–	–	–	–	64.5%	74.0%	72.7%
1990	61.6%	55.8%	51.2%	46	68.1%	74.4%	73.2%
1995	65.5%	54.5%	49.9%	50	71.9%	74.4%	73.4%
2000	68.6%	54.3%	49.8%	55	74.7%	74.8%	73.9%
2005	71.1%	52.8%	48.5%	39	77.0%	74.4%	73.6%
2010	72.7%	52.3%	47.9%	31	77.9%	74.0%	73.2%
2015	73.3%	51.1%	47.0%	34	78.2%	73.3%	72.7%
2020	67.3%	51.9%	47.7%	33	73.1%	74.4%	73.9%

The district rule marks a municipality as a core on its densely inhabited district population alone, using no commuting data. There is no 1985 Urban Employment Area delineation.

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The two cutoff anchors compared

Year	Cutoff	Commuting zones	Non-contiguous	Minimum contained	Mean contained	Share of the labour force	Areas split
2010	0.977	231	0	44.5%	90.0%	85.8%	31
2020	0.977	223	1	40.5%	90.4%	86.1%	33
2010	0.980	216	0	48.7%	90.8%	86.3%	28
2020	0.980	208	1	49.9%	90.8%	86.5%	32

Moving from 0.977 to 0.980 removes fifteen commuting zones in each year and lifts the minimum containment by four to nine points.

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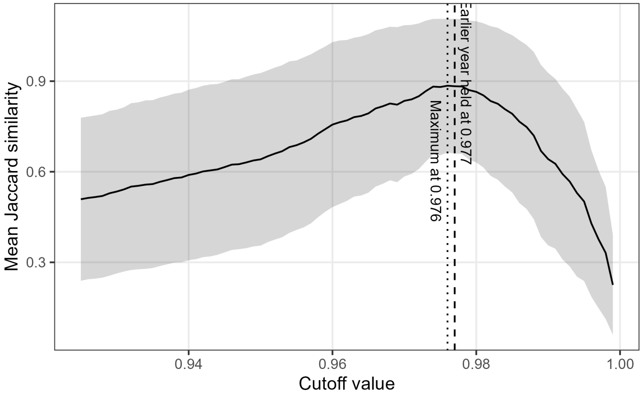
Every comparison pair

Pair	Role	Municipalities	Mean	Labour-force weighted
1980 against 1990	appendix	1,656	0.77	0.75
1990 against 2000	appendix	1,656	0.83	0.86
2000 against 2010	appendix	1,656	0.85	0.86
2005 against 2015	baseline	1,651	0.85	0.83
2010 against 2015	baseline	1,651	0.88	0.87
2010 against 2020	baseline	1,655	0.85	0.82

The 2010 to 2020 comparison sits below both pre-pandemic controls but above the decade pairs of the 1980s and 1990s.

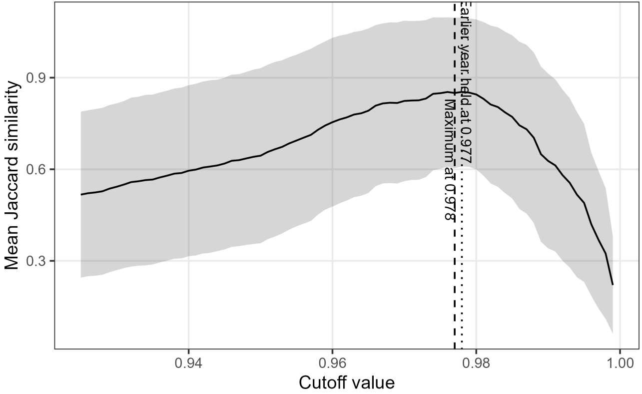
Control-pair curves Back

Similarity against the cutoff, 2010 against 2015



▸ 2005 against 2015 ◂ Back

Similarity against the cutoff, 2005 against 2015



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One departure from the existing pipeline

The existing Japanese pipeline joins the flow file to its own transpose in a way that keeps a pair only when the flow from i to j is recorded, so a pair connected only in the reverse direction enters as unconnected.

- The matrix it clusters is asymmetric; this replication uses the symmetric numerator the source method specifies.
- Between 1.2 and 4.4 percent of pairs are affected in each year, moving up to four zones and up to 108 municipalities.
- Measured year by year in `validation/notes/method_crosscheck.md`.

Where everything lives

On the repo: `https://github.com/daisukeadachi/commuting_zone_japan/`

- `validation/notes/replication_results.md` — the written results, with every number in this deck in context.
- `validation/notes/method_crosscheck.md` — the delineation formula across four implementations.
- `validation/notes/core_definition.md` — why the Urban Employment Area supplies the cores.
- `validation/output` — the tables behind every slide, as comma-separated files.
- `validation/code` — the nine scripts, run in the order they are listed in the results note.